

Math 142, F19
Quiz 13

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

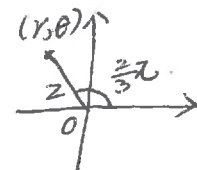
Problem 1. (6 points) The Cartesian coordinates of a point are given.

(a) $(-1, \sqrt{3})$. Find the polar coordinates (r, θ) of the point where $r > 0$ and $0 \leq \theta < 2\pi$.

$$r^2 = (-1)^2 + (\sqrt{3})^2 = 4 \Rightarrow r = 2 \text{ Since } r > 0$$

$$\left\{ \begin{array}{l} \tan \theta = \frac{\sqrt{3}}{-1} = -\sqrt{3}, \\ (-1, \sqrt{3}) \text{ is in the second quadrant} \end{array} \right. \Rightarrow \theta = \frac{2}{3}\pi$$

Since $0 \leq \theta < 2\pi$



$$(2, \frac{2}{3}\pi)$$

(b) $(3, -3)$. Find the polar coordinates (r, θ) of the point where $r < 0$ and $0 \leq \theta < 2\pi$.

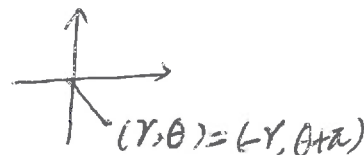
$$r^2 = 3^2 + (-3)^2 = 18 \Rightarrow r = \sqrt{18} = 3\sqrt{2} \text{ if } r > 0$$

$$\left\{ \begin{array}{l} \tan \theta = \frac{-3}{3} = -1 \\ (3, -3) \text{ lies in the fourth quadrant} \end{array} \right.$$

$\Rightarrow \theta = \frac{7}{4}\pi$ $\Rightarrow$ When $r < 0$,

$$\Rightarrow \theta = \frac{7}{4}\pi \Rightarrow \text{When } r < 0,$$

$$(-3\sqrt{2}, \frac{11}{4}\pi) = (-3\sqrt{2}, \frac{3}{4}\pi) \rightarrow 0 \leq \theta < 2\pi$$



Problem 2. (4 points) Identify the curve by finding a Cartesian equation for the curve described by the following polar equation:

$$r = 3 \sin \theta$$

$$r^2 = 3r \sin \theta$$

$$x^2 + y^2 = 3y$$

$$x^2 + y^2 - 3y = 0$$

$$x^2 + (y - \frac{3}{2})^2 = (\frac{3}{2})^2$$

$\Rightarrow$ A circle centered at $(0, \frac{3}{2})$ with radius $\frac{3}{2}$.